

$^9\text{Be}(^{36}\text{Si}, ^{35}\text{Al})\gamma$ 2014St18

$^{36}\text{Si} \rightarrow \text{p} + ^{35}\text{Al}$ from $J^\pi = 0^+$ ^{36}Si ground state.

2014St18: A ^{36}Si secondary beam was produced via the projectile fragmentation of a 140-MeV/nucleon ^{48}Ca primary beam impinging on a ^9Be target at NSCL, MSU and was selected by the A1900 separator. The states of ^{35}Al and ^{35}Si were populated by the one-proton/neutron knockout reactions, respectively, from the ^{36}Si beam at a midtarget energy of 97.7 MeV/nucleon 5 on a 287-mg/cm² ^9Be secondary target. Knockout residues were identified from their energy loss measured by an ionization chamber at the focal plane of the S800 spectrometer and from their ToF measured between two scintillators at the object position and at the focal plane of the S800 spectrometer. The position and angle of the residues were measured using two cathode-readout drift chambers. Prompt γ rays from the deexcitation of the residues were detected by the GRETINA Ge array. Measured Doppler-corrected E_γ , I_γ , $(^{35}\text{Al})\gamma$ -coin, $\gamma\gamma$ -coin, the parallel momentum distributions of populated states in ^{35}Al residues. Deduced levels, J , π , L-transfers, inclusive and exclusive knockout cross section for producing ^{35}Al from ^{36}Si . Calculations using eikonal model and shell model calculations with SDPF-U and SDPF-MU interactions.

 ^{35}Al Levels

Total knockout $\sigma = 22$ mb / for producing ^{35}Al from ^{36}Si .

$E(\text{level})^\dagger$	$J^\pi^\ddagger$	$L^\ddagger$	Comments
0	$(5/2)^+$	2	J^π : interpreted as the $1d_{5/2}$ proton removal from ^{36}Si (2014St18). Partial knockout $\sigma = 13$ mb 2.
801 3			Partial knockout $\sigma = 1.0$ mb 7.
1005 3			Partial knockout $\sigma = 0.8$ mb 9.
1865 4			Partial knockout $\sigma = 1.0$ mb 2.
1973 4	$3/2^+, 5/2^+$	2	Partial knockout $\sigma = 3.2$ mb 5.
2733 7			Partial knockout $\sigma = 0.5$ mb 1.
3244 5	$3/2^+, 5/2^+$	2	Partial knockout $\sigma = 2.6$ mb 3.
4275? 9	$3/2^+, 5/2^+$	2	Partial knockout $\sigma = 0.5$ mb 1.

[†] From a least-squares fit to γ -ray energies.

[‡] **2014St18** deduced L by comparing the measured and eikonal-calculated parallel momentum distributions of residuals. J^π options are deduced accordingly.

 $\gamma(^{35}\text{Al})$

E_γ	I_γ	$E_i(\text{level})$	J_i^π	E_f	J_f^π	Comments
802 4	10 1	801		0	$(5/2)^+$	
859 4	3.6 3	1865		1005		
968 4	4.4 3	1973	$3/2^+, 5/2^+$	1005		
1003 4	19 1	1005		0	$(5/2)^+$	
1064 4	0.8 2	1865		801		
1174 5	2.8 3	1973	$3/2^+, 5/2^+$	801		
^x 1473 5	1.1 2					
1932 6	2.5 3	2733		801		
1972 6	7.5 5	1973	$3/2^+, 5/2^+$	0	$(5/2)^+$	
2237 6	7.8 6	3244	$3/2^+, 5/2^+$	1005		
2440 7	1.4 2	3244	$3/2^+, 5/2^+$	801		
^x 3060 8	1.6 4					
3250 8	3.3 4	3244	$3/2^+, 5/2^+$	0	$(5/2)^+$	
4275 [†] 9	3 1	4275?	$3/2^+, 5/2^+$	0	$(5/2)^+$	2014St18 states that the 4275 γ transition could populate the ground state or one of the lowest two excited states. In the absence of any observed $\gamma\gamma$ coincidences, they tentatively place the 4275-keV γ transition feeding the ground state.

Continued on next page (footnotes at end of table)

$^9\text{Be}(^{36}\text{Si}, ^{35}\text{Al}\gamma)$ 2014St18 (continued) $\gamma(^{35}\text{Al})$ (continued)[†] Placement of transition in the level scheme is uncertain.^x γ ray not placed in level scheme. $^9\text{Be}(^{36}\text{Si}, ^{35}\text{Al}\gamma)$ 2014St18

Level Scheme

Intensities: Yield/100 ions

Legend

- $I_\gamma < 2\% \times I_\gamma^{\max}$
- $I_\gamma < 10\% \times I_\gamma^{\max}$
- $I_\gamma > 10\% \times I_\gamma^{\max}$
- - - → γ Decay (Uncertain)
- Coincidence

